RAM-Like, High Density, Field-Free 2-Terminal SOT-MRAM for Energy-Efficient AI Compute at Nanosecond Timescale

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Energy-efficient computing is critical for sustaining the rapid growth of AI workloads driven by large language models. Two-terminal Spin Orbit Torque Magnetoresistive Random Access Memory (2-terminal SOT-MRAM) offers a promising solution by combining high cell density, low power consumption, and nanosecond-scale performance, *without the need of an external magnetic field*. Compared with conventional STT-MRAM, SOT architecture reduces switching current density and enables significantly faster switching. Experimental results show that y-type anisotropy 2-terminal SOT-MRAM devices achieve more efficient SOT-driven switching, lower write voltages, and faster settling times than z-type devices. These advantages position 2-terminal SOT-MRAM as a strong candidate for next-generation energy-efficient AI memory and high-performance computing systems.

Index Terms—**RAM, MRAM, SOT-MRAM, Artificial Intelligence, AI, spintronics, memory**

I. Introduction

Energy-efficient computing, particularly in exploding data-intensive AI applications spurred by large language models (LLM), is essential for addressing the rising power demands and ensuring sustainable technological advancement. Spin Orbit Torque Magnetoresistive Random Access Memory (SOT-MRAM) has emerged as a promising technology in this domain, offering either embedded nonvolatile memory (eNVM) or RAM-like solutions such as last level cache (LLC). In particular, RAM-like, high density, field-free 2-terminal SOT-MRAM combines low power consumption with high performance, and promises higher bit-cell densities than CMOS SRAM or 3-terminal SOT-MRAM [1-3] at deep technology nodes (e.g., N5 or below). In this work, we show that next generation MRAM technologies (**Fig. 1**), specifically 2-terminal SOT-MRAM, are well-suited for energy-efficient AI compute at nanosecond timescale.

II. Experimental Results

Two-terminal SOT-MRAM configuration (**Fig. 1b**) is similar to STT-MRAM (**Fig. 1a**), except an SOT wire is placed beneath the free layer of the MTJ. The advantages of the SOT wire are two-fold. First, the critical switching current density is reduced because additional SOT spins are generated by the SOT material, reducing the switching voltage. Second, SOT switching is shown to be significantly faster than STT switching due to its switching physics [4,5].

In comparing our fabricated devices with y-type and z-type magnetic anisotropies, we find that the y-type device allows for SOT-dominated switching at smaller critical dimensions, which results in faster settling times over the z-type SOT or STT devices. The z-type device allows for a smaller device footprint at the same technology node due to its interfacial perpendicular magnetic anisotropy but requires higher SOT current densities than the y-type SOT devices. The y-type SOT-MRAM makes the most efficient use of SOT spins, resulting in lower write voltages or currents because the SOT along y-axis can be synergistic with STT torque. The z-type SOT-MRAM has the SOT along y-axis and STT along z-axis, together they can achieve deterministic switching without an external field. Both types of SOT-MRAM devices were switched (field-free) at current pulse widths ranging from 10 µs to 2.5 ns (**Fig. 2**). The actual critical switching current densities are also greatly affected by the SOT and STT efficiencies of the materials used.

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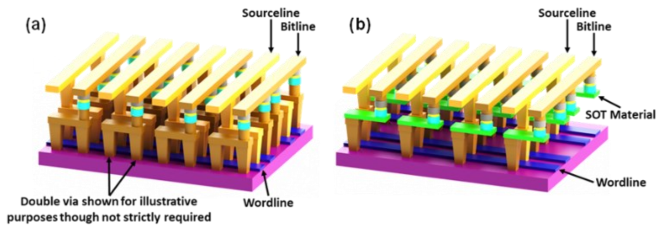

Fig. 1. (a) Typical STT-MRAM; (b) two-terminal SOT-MRAM with the SOT material below the MTJ. Both are known to have higher cell densities than 3-terminal SOT-MRAM or SRAM.

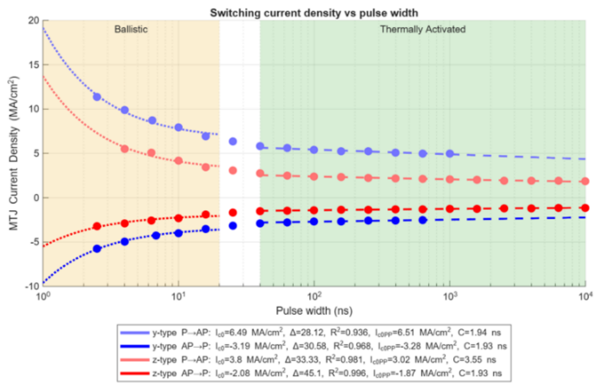

Fig. 2. Critical switching current density vs. pulse width for y-type 2-terminal SOT-MRAM device (red) and z-type 2-terminal SOT-MRAM device (blue) for the ballistic switching region (<< 20 ns) and thermally activated region (> 40 ns). Measured data was limited to 1.8V maximum to avoid damaging the devices. For the z-type device (blue), the characteristic switching time constant was 4.2ns (P-AP) and 2.2ns (AP-P). For the y-type device (red), it was 2.2ns (P-AP and AP-P). The dotted lines are fit curves using the ballistic switching model from, revealing the high-speed strengths of the y-type 2-terminal SOT-MRAM. [5] (AP = Antiparallel, P = Parallel)